

Hierarchical Equal Risk Contribution Is a Corner of a Schur Square, and Two Dials Lead to Minimum Variance

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Abstract

Hierarchical equal risk contribution cuts a dendrogram into clusters, holds the inverse-variance portfolio inside each cluster, and budgets across clusters by inverse cluster variance. We show that its budget rule is already the one the global minimum-variance portfolio uses, and that what it lacks is conditioning in two places: each cluster on the other clusters, and each asset on its cluster mates. Damping the two Schur complements by γ and η gives a square with the method at one corner and the global optimum at the opposite corner. The inner dial has a closed form that interpolates the naive inverse-variance quantities and Stevens' regression quantities, keeps every weight positive up to an explicit frontier, and on an equicorrelated cluster traces the straight segment from inverse variance to the cluster's minimum-variance portfolio. Under estimation error the best point can be strictly inside the square, and we give the two-dial local theorem at the minimum-variance corner, a sign criterion for the inner dial with both signs realized, and exact examples, including one whose optimum is $\eta = 3/5$ with gaps $1/45$ and $1/20$ to the two ends.

1 Introduction

Hierarchical equal risk contribution, HERC [1], is a two-level rule. A dendrogram is cut into k clusters. Inside each cluster the weights are inverse variance. Across clusters the budgets follow the inverse of each cluster's risk. The method is widely implemented [14, 15, 16] and widely tested [11, 12, 13], with results that are mixed and that no theory has explained.

Block inversion gives the global minimum-variance portfolio the same two-level shape. Each cluster's block of $\Sigma^{-1}\mathbf{1}$ is a minimum-variance direction on the cluster's covariance conditional on everything else, and the cluster totals are whatever those directions sum to [3]. Written as a budget rule, the total is the inverse of the cluster's conditional minimum variance. That is HERC's rule.

So HERC's across-cluster step is not the approximation. The approximations are that its inside step reads a diagonal, and that neither step reads anything outside the cluster. Both are the omission of a Schur complement. Damping the two complements by separate parameters gives a square. HERC sits at $(\gamma, \eta) = (0, 0)$ and the global minimum-variance portfolio at $(1, 1)$.

The inner dial is new and simple. A single asset conditioned on its cluster mates is Stevens' regression identity [6], and damping it gives a weight whose numerator and denominator are each linear in η between the naive quantity and the regression quantity. The whole inner dial costs one

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inverse per cluster. The outer dial is the flat bridge of [5], with the outer optimizer replaced by HERC’s inverse fitness.

Section 6 asks where to sit when the covariance is estimated. The inner dial on an equicorrelated cluster reduces to one scalar, which gives exact interior optima and a sign criterion at the full-coupling end. At the minimum-variance corner the two-dial local theorem says which dials move inside and by how much, and exact examples show both dials moving, and one moving while the other stays.

Everything exact here concerns the unconstrained, fully invested, long-short problem, as in [3, 4]. HERC is long-only by construction, and Proposition 4 says exactly how far the inner dial can be turned before that is lost.

2 The method

Let $\Sigma \succ 0$ be the covariance of n assets and let $C_1, \dots, C_k$ be the clusters at which HERC cuts its dendrogram. For an index set C write Σ_{CC} for its block, $-C$ for its complement, and σ_i^2 for Σ_{ii} .

HERC holds inside cluster C the inverse-variance portfolio $v_C \propto \text{diag}(\Sigma_{CC})^{-1}\mathbf{1}$, normalized to $\mathbf{1}^\top v_C = 1$, and measures the cluster’s risk by that portfolio’s variance, $\nu_C = v_C^\top \Sigma_{CC} v_C$. It then budgets across clusters top down along the dendrogram.

The split rule. Raffinot writes the share of the first child at a split as $\text{RC}_1/(\text{RC}_1 + \text{RC}_2)$, “where RC_1 is the risk contribution of the first cluster” [1], and does not define RC . Read as a risk, the formula gives the riskier child more. Read as an inverse risk, a precision, it gives the riskier child less, which is the intent, and Palomar’s textbook makes the same reading in recovering the inverse-variance split [8]. We take RC of a child to be the sum of the precisions $1/\nu_C$ of the final clusters it contains.

Proposition 1 (The dendrogram telescopes). *With additive precisions, top-down division along any dendrogram gives final cluster C the budget $a_C \propto 1/\nu_C$. The tree affects the allocation only through the choice of clusters.*

Proof. At a node with children L and R and precisions $P_L = \sum_{C \subseteq L} 1/\nu_C$ and P_R , the child L receives the fraction $P_L/(P_L + P_R)$ of the node’s budget. Along the path from the root to C the fractions telescope to $(1/\nu_C)/P_{\text{root}}$. $\square$

So HERC with variance is the flat two-level rule

$$w \propto \sum_C \frac{v_C}{\nu_C}, \quad v_C \propto \text{diag}(\Sigma_{CC})^{-1}\mathbf{1}, \quad \nu_C = v_C^\top \Sigma_{CC} v_C. \quad (1)$$

In words: hold inverse variance inside each cluster, and give each cluster capital in proportion to the inverse variance of what it holds.

Remark 1 (The reference implementations). *The libraries [14, 15, 16] implement the split as one minus the ratio of the children’s risks, with the risk of a child taken as the sum of the variances ν_C of its final clusters. For two clusters this agrees with (1). For three or more it does not, because sums of variances do not telescope, and the allocation then depends on the shape of the dendrogram above the cut. The verification script exhibits both facts. We analyze (1).*

3 Block inversion

Fix a vector u and a cluster C . Block inversion of Σ on the partition $(C, -C)$ gives [3, App. A]

$$(\Sigma^{-1}u)_C = (\Sigma_C^c)^{-1}b_C, \quad \Sigma_C^c = \Sigma_{CC} - \Sigma_{C,-C}\Sigma_{-C,-C}^{-1}\Sigma_{-C,C}, \quad b_C = u_C - \Sigma_{C,-C}\Sigma_{-C,-C}^{-1}u_{-C}. \quad (2)$$

The complement is the covariance of C conditional on the other assets and b_C is the companion vector. For positive definite Q and a vector b the constrained problem of minimizing $x^\top Qx$ subject to $b^\top x = 1$ has solution and attained variance

$$v(Q, b) = \frac{Q^{-1}b}{b^\top Q^{-1}b}, \quad \nu(Q, b) = \frac{1}{b^\top Q^{-1}b}, \quad \text{so} \quad Q^{-1}b = \frac{v(Q, b)}{\nu(Q, b)}. \quad (3)$$

The unnormalized direction is the normalized portfolio divided by its variance.

Apply this to (2) with $u = \mathbf{1}$. The global minimum-variance portfolio is $\sum_C v_C / \nu_C$ with $v_C = v(\Sigma_C^c, b_C)$ and $\nu_C = \nu(\Sigma_C^c, b_C)$, which is (1) with two substitutions undone. HERC replaces the conditional pair (Σ_C^c, b_C) by the raw pair $(\Sigma_{CC}, \mathbf{1})$, and it replaces the minimum-variance portfolio of that pair by the inverse-variance portfolio. The budget rule is untouched.

Noguer i Alonso lists three substitutions that turn the exact bisection identity into hierarchical risk parity [2, 7]. HERC makes the first two, the raw block for the complement and $\mathbf{1}$ for b , together with the diagonal for the block inside each cluster. It does not make the third, because its inter-cluster budget is the exact one.

One asset at a time. Take $C = \{i\}$ in (2) with $u = \mathbf{1}$. Then $\Sigma_{\{i\}}^c$ is the residual variance of asset i regressed on all the others and $b_{\{i\}}$ is one minus the sum of the regression coefficients. This is Stevens' identity [6],

$$(\Sigma^{-1}\mathbf{1})_i = \frac{1 - \sum_{j \neq i} \beta_{ij}}{\sigma_i^2 (1 - R_i^2)}, \quad (4)$$

with R_i^2 the multiple correlation and β_{ij} the hedge coefficients. The inverse-variance portfolio is the same expression with the hedges set to zero. Inside a cluster, HERC is therefore Stevens with the regression switched off.

4 The square

Two Schur complements are missing from (1): that of a cluster against the other clusters, and that of an asset against its cluster mates. Damp each.

Definition 1 (The two dials). *For $\gamma \in [0, 1]$ the cluster pair of C is*

$$Q_C(\gamma) = \Sigma_{CC} - \gamma \Sigma_{C,-C} \Sigma_{-C,-C}^{-1} \Sigma_{-C,C}, \quad b_C(\gamma) = u_C - \gamma \Sigma_{C,-C} \Sigma_{-C,-C}^{-1} u_{-C}. \quad (5)$$

For $\eta \in [0, 1]$ and a pair (Q, b) on C , the leaf direction $z(Q, b, \eta)$ has entries

$$z_i = \frac{b_i - \eta Q_{i,-i} Q_{-i,-i}^{-1} b_{-i}}{Q_{ii} - \eta Q_{i,-i} Q_{-i,-i}^{-1} Q_{-i,i}}, \quad i \in C, \quad (6)$$

where $-i$ now means $C \setminus \{i\}$. The portfolio at (γ, η) is

$$w_{\gamma\eta} \propto \sum_C \frac{b_C^\top z_C}{z_C^\top Q_C z_C} z_C, \quad z_C = z(Q_C(\gamma), b_C(\gamma), \eta), \quad (7)$$

with a cluster whose direction z_C vanishes contributing nothing.

When $b_C^\top z_C \neq 0$ the summand is v_C/ν_C with $v_C = z_C/b_C^\top z_C$ the cluster's holding normalized by its companion vector and $\nu_C = v_C^\top Q_C v_C$ its variance on the pair, so (7) is the inverse-variance budget rule in the form (3). The unnormalized form is the one we use, because it has no singularity. On ordinary positive-definite input a companion vector can vanish: $\Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ with singleton clusters has $b_{\{1\}}(1) = 0$, so $z_{\{1\}} = 0$ at $\gamma = 1$ and the normalized holding $v_{\{1\}}$ is 0/0, while $\Sigma^{-1}\mathbf{1} = (0, 1)^\top$ is perfectly well defined. By (2) a vanishing companion means that block of the global direction is zero, which is what (7) assigns. The same convention is used for the flat bridge in [5].

The dial γ conditions a cluster on the outside world. The dial η conditions an asset on its cluster mates. The budget rule is HERC's own, the inverse variance of what the cluster holds, measured on the cluster's pair. We take $u = \mathbf{1}$ unless said otherwise.

Proposition 2 (The four corners).

1. $(0, 0)$ is HERC, (1) .
2. $(1, 1)$ is the global direction $\Sigma^{-1}u$, so with $u = \mathbf{1}$ the global minimum-variance portfolio.
3. $(0, 1)$ holds the minimum-variance portfolio of each cluster's own block and budgets by inverse cluster minimum variance.
4. $(1, 0)$ holds the inverse-variance portfolio of each cluster's conditional pair, with the companion vector in the numerator, and budgets by its inverse variance on the pair.

Proof. At $\eta = 0$ the leaf direction is $z_i = b_i/Q_{ii}$, which at $\gamma = 0$ is $1/\sigma_i^2$. At $\eta = 1$ the leaf direction is the block inversion (2) of the pair (Q, b) on the partition $(\{i\}, C \setminus \{i\})$, so $z = Q^{-1}b$, and then $b^\top z / (z^\top Q z) = 1$, so the summand is $Q^{-1}b$ whether or not b vanishes. At $\gamma = 1$ the pair is $(\Sigma_C^\mathcal{C}, b_C)$ and $Q^{-1}b = (\Sigma^{-1}u)_C$ by (2). The remaining corners read off the definitions. $\square$

corner	inside a cluster	across clusters
$(0, 0)$ HERC	inverse variance on Σ_{CC}	inverse variance of the holding
$(0, 1)$	minimum variance on Σ_{CC}	inverse minimum variance
$(1, 0)$	inverse variance on the conditional pair	inverse variance of the holding
$(1, 1)$ global	minimum variance on the conditional pair	inverse conditional minimum variance

Table 1: The four corners with $u = \mathbf{1}$. The across-cluster rule is the same at every corner; only what it measures changes.

The corner $(0, 1)$ is a cluster-then-optimize rule with inverse-variance budgets, of the kind used in [10] with a different inner objective. The corner $(1, 0)$ has no name. Along the edge $\eta = 1$ the square is the flat bridge of [5] with the outer optimizer replaced by HERC's inverse fitness; the two agree at $\gamma = 1$ and differ inside.

Remark 2 (Cheaper conditioning sets). *The pair (5) conditions on all $n - |C|$ other assets, which is the inverse a two-level method exists to avoid. Two devices from earlier work apply unchanged. Conditioning composed top down along the dendrogram, with each block conditioned on its sibling as in [3], reaches the same pair at $\gamma = 1$ because Schur complements compose, and the script checks this on an unbalanced tree. Conditioning on one knot per other cluster, as in [5], is exact under the gateway model stated there. The tree device inverts sibling blocks, and the knot device inverts only cluster blocks and one $k \times k$ matrix.*

Remark 3 (Other companion vectors). *Proposition 2 holds for any u . With $u = \sigma$, the vector of volatilities, the corner $(1, 1)$ is the most diversified portfolio $\Sigma^{-1}\sigma$ [9], and the corner $(0, 0)$ holds the inverse-volatility portfolio inside each cluster with the cash budget of cluster C proportional to $\text{DR}_C^2/\bar{\sigma}_C$, where DR_C is the diversification ratio of the holding and $\bar{\sigma}_C$ its average volatility. HERC with volatility as the risk measure, as implemented in [14], holds the same inside portfolio but budgets by $\text{DR}_C/\bar{\sigma}_C$. It is not a corner of this square, and the diversification ratio enters squared in the rule that is. With $u = \mu$ the corner $(1, 1)$ is the tangency direction, and $(0, 0)$ holds μ_i/σ_i^2 inside each cluster with budgets proportional to the squared Sharpe ratio of the holding divided by its mean.*

5 The inner dial

The leaf direction (6) looks like $|C|$ solves of size $|C| - 1$. It is one solve.

Proposition 3 (Closed form). *Let $q_i = 1/(Q^{-1})_{ii}$ and $s_i = (Q^{-1}b)_i/(Q^{-1})_{ii}$. Then*

$$z_i(\eta) = \frac{(1 - \eta) b_i + \eta s_i}{(1 - \eta) Q_{ii} + \eta q_i}. \quad (8)$$

Proof. Block inversion on $(\{i\}, -i)$ gives $Q_{ii} - Q_{i,-i}Q_{-i,-i}^{-1}Q_{-i,i} = 1/(Q^{-1})_{ii}$ and $b_i - Q_{i,-i}Q_{-i,-i}^{-1}b_{-i} = (Q^{-1}b)_i/(Q^{-1})_{ii}$. Both expressions in (6) are affine in η between their values at 0 and 1. $\square$

At $\eta = 1$ the weight is $(Q^{-1}b)_i$, and with $b = \mathbf{1}$ the numbers q_i and s_i are Stevens' residual variance and one minus the hedge sum (4). The dial interpolates both the numerator and the denominator of Stevens' formula linearly. The denominator is a convex combination of two positive numbers, so it never vanishes.

Proposition 4 (Long-only frontier). *Let $b > 0$ entrywise. The leaf direction $z(\eta)$ is nonnegative in every entry if and only if $\eta \leq \eta_+$, where*

$$\eta_+ = \min_{i: s_i < 0} \frac{b_i}{b_i - s_i}, \quad (9)$$

and $\eta_+ = 1$ when no s_i is negative. The first short appears at the asset whose hedge sum first exceeds its companion entry.

Proof. The denominator in (8) is positive. The numerator is affine in η , equals $b_i > 0$ at $\eta = 0$, and is negative at $\eta = 1$ exactly when $s_i < 0$, crossing zero at $\eta = b_i/(b_i - s_i)$. $\square$

HERC is long-only by construction and the global minimum-variance portfolio is not. The frontier says how much of the inside conditioning can be restored before that changes, cluster by cluster, at the cost of the inverse already computed.

The equicorrelated cluster. Let cluster C have m assets with volatilities σ_i and a common correlation ρ , and let $\gamma = 0$ so the pair is $(\Sigma_{CC}, \mathbf{1})$. Write $\kappa = \rho/(1 + (m - 2)\rho)$ for the regression coefficient of one standardized asset on another, and $s_{-i} = \sum_{j \neq i} 1/\sigma_j$.

Proposition 5 (A straight segment). *On an equicorrelated cluster,*

$$z_i(\eta) \propto \frac{1 - \eta \kappa \sigma_i s_{-i}}{\sigma_i^2}, \quad (10)$$

with a factor common to all i . The normalized holding lies on the straight segment from the inverse-variance portfolio v_0 to the cluster's minimum-variance portfolio v_* ,

$$v(\eta) = v_* + (1 - \theta(\eta\kappa))(v_0 - v_*), \quad \theta(t) = \frac{t(\alpha - \kappa\beta)}{\kappa(\alpha - t\beta)}, \quad (11)$$

where $\alpha = \sum_i \sigma_i^{-2}$ and $\beta = \sum_i s_{-i}/\sigma_i$, so that $\beta = (\sum_i \sigma_i^{-1})^2 - \alpha$. With $h_i = \kappa\sigma_i s_{-i}$ the long-only frontier is

$$\eta_+ = \min(\{1\} \cup \{1/h_i : h_i > 1\}), \quad (12)$$

which is $\min_i 1/h_i$ when some asset is shorted inside the square and 1 otherwise, including every $\rho \leq 0$, where $\kappa \leq 0$ and nothing is ever shorted.

Proof. In the equicorrelated model every asset has the same multiple correlation on the rest, so the denominators in (6) share one factor, and the hedge sum of asset i is $\kappa\sigma_i s_{-i}$. So the unnormalized weight is $a_i - t c_i$ with $t = \eta\kappa$, $a_i = \sigma_i^{-2}$ and $c_i = s_{-i}/\sigma_i$. Two normalized vectors of this form differ by a multiple of the fixed vector $a\beta - c\alpha$, which gives the segment, and $t = \kappa$ is the minimum-variance end. The frontier is (9) with $b = \mathbf{1}$ and $s_i = 1 - h_i$: the numerator of asset i crosses zero inside $[0, 1]$ exactly when $h_i > 1$, at $\eta = 1/h_i$. $\square$

In words: on an equicorrelated cluster, the inner dial is linear shrinkage between the two familiar portfolios, with the mixing weight (11) a fixed function of the estimated correlation. With equal volatilities the two ends coincide and the dial does nothing.

The cut. With one cluster, γ is idle and the square is the inner dial applied to all of Σ , from inverse variance to global minimum variance. With n singleton clusters, η is idle and the pair of $\{i\}$ is Stevens' pair, so the outer dial traces the same path. HERC is the inverse-variance portfolio at both extremes of the cut, and departs from it only through the budgets (1) at intermediate cuts. The dendrogram chooses where on that path the two dials separate.

6 Where to sit

Let $\hat{\Sigma}_\tau = \Sigma + \tau E$ be an estimate, where E is a random symmetric matrix whose law is unchanged by $E \mapsto -E$, and $\hat{\Sigma}_\tau \succ 0$ on the range of τ considered. The clusters are held fixed. Every quantity in (7) is computed from $\hat{\Sigma}_\tau$ and normalized by $\mathbf{1}^\top w = 1$. Define

$$F(\gamma, \eta, \tau) = \mathbb{E}[w_{\gamma\eta}(\hat{\Sigma}_\tau)^\top \Sigma w_{\gamma\eta}(\hat{\Sigma}_\tau)], \quad V_0 = F(\cdot, \cdot, 0), \quad G = \frac{1}{2} \partial_\tau^2 F(\cdot, \cdot, 0), \quad (13)$$

and assume the expansion $F = V_0 + \tau^2 G + O(\tau^4)$ holds in C^2 in the dials, as in [5].

Lemma 1 (The HERC edge is blind). *For every η , the portfolio $w_{0\eta}$ depends on $\hat{\Sigma}_\tau$ only through its diagonal blocks $\hat{\Sigma}_{CC}$.*

Proof. At $\gamma = 0$ the pair of C is $(\hat{\Sigma}_{CC}, \mathbf{1})$, the leaf direction and the fitness read only the pair, and the budget rule reads only the fitnesses. $\square$

So under noise supported off the diagonal blocks, $F(0, \eta, \tau) = V_0(0, \eta)$ exactly, and the argument of [4] transfers: the cost of turning γ away from zero is of order $\gamma\tau^2$, and at any η with $\partial_\gamma V_0(0, \eta) < 0$ the point $(0, \eta)$ is strictly suboptimal for small τ . Under whole-matrix noise the edge is itself noisy and the comparison is between noise costs, as discussed there.

Proposition 6 (The minimum-variance corner, two dials). *Let $J = [\partial_\gamma w, \partial_\eta w]$ at $(1, 1, 0)$ and suppose its columns are linearly independent. Then $\nabla V_0(1, 1) = 0$, the Hessian is $H = 2J^\top \Sigma J \succ 0$, and for all sufficiently small τ the map $F(\cdot, \cdot, \tau)$ has a unique stationary point near $(1, 1)$, a strict local minimizer, at*

$$(\tilde{\gamma}, \tilde{\eta}) = (1, 1) - H^{-1} \nabla G(1, 1) \tau^2 + O(\tau^4). \quad (14)$$

If both components of $H^{-1} \nabla G(1, 1)$ are positive the local minimizer is strictly inside the square for small $\tau > 0$. If one is negative the stationary point lies outside the square, the constrained minimizer lies on one of the two edges through the corner, and its position along an edge is given by the one-dial formula with the edge's own second derivatives.

Proof. Since $\mathbf{1}^\top w = 1$, every derivative of w in the dials has zero sum. By Proposition 2, $\Sigma w_{11} \propto \mathbf{1}$, so first derivatives of V_0 vanish and in the second derivatives the terms containing second derivatives of w vanish, leaving $2J^\top \Sigma J$. The implicit function theorem applied to $\nabla_{\gamma\eta} F = 0$ at $(1, 1, 0)$ gives (14) and strict convexity on a neighbourhood not depending on τ . The edge statement is the one-dial case of the same argument. $\square$

The displacement is of second order in the noise and the gain from making it is of fourth order, as in [5]. The structure of the square does not fix the signs; both occur below.

6.1 The inner dial, exactly

Let $\gamma = 0$ and take a single equicorrelated cluster as in Proposition 5, with the volatilities known and the correlation estimated, $\hat{\rho} = \rho + \tau Z$ for a symmetric Z . Then the recipe holds $v(\eta)$ at $t = \eta \kappa(\hat{\rho})$, while the true portfolio that minimizes variance is at $t = \kappa(\rho)$. By (11) and the segment,

$$F(\eta, \tau) = V_\star + \Delta \mathbb{E}[(1 - \theta(\eta \kappa(\hat{\rho})))^2], \quad 1 - \theta(t) = \frac{(\kappa - t) \alpha}{\kappa(\alpha - t\beta)}, \quad (15)$$

where $V_\star$ is the cluster's minimum variance and $\Delta = V(v_0) - V_\star$ is the excess variance of inverse variance over it. The inner dial's noise problem is one scalar, and nothing else about the cluster enters.

An exact interior optimum. Let the estimate be $\hat{\rho} \in \{0, 2\rho\}$ with equal probability. On the branch $\hat{\rho} = 0$ the dial does nothing. On the branch $\hat{\rho} = 2\rho$ the choice

$$\eta_\star = \frac{\kappa(\rho)}{\kappa(2\rho)} = \frac{1 + 2(m-2)\rho}{2(1 + (m-2)\rho)} \quad (16)$$

puts t at $\kappa(\rho)$ and hence $\theta = 1$, the population optimum. One branch is constant and the other attains its minimum, uniquely because θ is strictly monotone in t . So $\eta_\star$ is the unique global optimum, strictly inside, independent of the volatilities. With three assets of volatilities $(1, 2, 2)$ and $\rho = \frac{1}{4}$ the optimum is $\eta_\star = \frac{3}{5}$, and in exact arithmetic $F(0) - F(\frac{3}{5}) = \frac{1}{45}$ and $F(1) - F(\frac{3}{5}) = \frac{1}{20}$.

Proposition 7 (Sign at the full-coupling end). *Let $\hat{\rho} = \rho \pm \tau$ with equal probability, $q = (m-2)\rho$, and $r = \beta/\alpha$. Then $\partial_\eta F(1, \tau)$ has the sign of*

$$\alpha(1 - q) + \kappa\beta(2 + q), \quad \text{equivalently} \quad q^2 - 1 < r\rho(2 + q), \quad (17)$$

for all sufficiently small τ , and the optimum is interior when this holds and stays at $\eta = 1$ when the reverse strict inequality holds. The interior optimum sits at

$$\tilde{\eta} = 1 - \frac{\alpha(1 - q) + \kappa\beta(2 + q)}{\rho^2(1 + q)^2(\alpha - \kappa\beta)} \tau^2 + O(\tau^4). \quad (18)$$

These statements are global on $[0, 1]$, since V_0 has the unique minimizer $\eta = 1$ and $F(\cdot, \tau) \rightarrow V_0$ uniformly.

Proof. Differentiate (15) at $\eta = 1$: $\partial_\eta F(1, \tau) = 2\Delta\alpha^2(\alpha - \kappa\beta)\kappa^{-2} \mathbb{E}[\varphi(\kappa(\hat{\rho}))]$ with $\varphi(x) = x(x - \kappa)/(\alpha - x\beta)^3$. Expand $\mathbb{E}[\varphi(\kappa(\rho \pm \tau))]$ to second order using $\varphi(\kappa) = 0$, $\varphi'(\kappa) = \kappa/(\alpha - \kappa\beta)^3$, $\varphi''(\kappa) = 2(\alpha + 2\kappa\beta)/(\alpha - \kappa\beta)^4$, $\kappa' = (1 + q)^{-2}$ and $\kappa\kappa'' = -2q(1 + q)^{-4}$. The bracket collects to $[\alpha(1 - q) + \kappa\beta(2 + q)]/(1 + q)^4$ over $(\alpha - \kappa\beta)^4$. Dividing by $V_0''(1) = 2\Delta\alpha^2/(\alpha - \kappa\beta)^2$ gives (18). Multiplying (17) by $(1 + q)/\alpha$ gives the second form. $\square$

In words: the dial should stop short of full coupling unless the cluster is both large and highly correlated, with $(m - 2)\rho$ well above one, and its precision is concentrated in one asset, so that r is small. Equal volatilities give $r = m - 1$, which always satisfies (17), but there the two ends coincide. The script confirms both signs: volatilities $(1, 2, 3, 5)$ with $\rho = \frac{1}{10}$ move inside, and volatilities $(1, 20, 30, 50)$ with $\rho = \frac{9}{10}$ stay at full coupling.

Why the sign can go either way. The mixing weight $\theta(\eta\kappa(\hat{\rho}))$ is a nonlinear function of the estimated correlation, so symmetric noise in $\hat{\rho}$ shifts its mean as well as spreading it, and the direction of the shift depends on the cluster. Damping by η shrinks the spread, which always helps, and moves the mean, which helps or hurts according to that direction. The sign of (17) is the outcome of the competition.

6.2 Both dials

Take two clusters of two assets. Each has a unit-variance knot p and a member m of volatility s correlated r with its knot, the knots have correlation c , and the members' cross-cluster covariance passes through the knots, so that $\text{Cov}(m_1, p_2) = r_1 s_1 c$ and $\text{Cov}(m_1, m_2) = r_1 s_1 r_2 s_2 c$. Let all three correlations be estimated as $r_1 \pm \tau$, $r_2 \pm \tau$, $c \pm \tau$ independently. The verification script computes the derivatives in Proposition 6 as truncated power series with rational coefficients, so (14) is exact.

With $(r_1, r_2, c, s_1, s_2) = (\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, 2, 3)$ both shifts are negative, about $-29.9\tau^2$ in γ and $-11.8\tau^2$ in η , so the optimum is strictly inside the square. With $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, \frac{1}{2}, 3)$ the γ shift is positive, about $+15.1\tau^2$, and the η shift negative, so the optimum sits on the edge $\gamma = 1$ at $\eta \approx 1 - 16.6\tau^2$ by the edge formula. At $\tau = \frac{1}{40}$ the predicted point beats the corner in both examples in exact arithmetic. Noise in the cross-cluster correlation alone still moves η , through the mixed term of G , so the two dials are not separable in general.

7 Cost

At $\gamma = 0$ the square costs one inverse of each cluster block, from which (8) gives the whole inner dial and the fitness. The outer dial costs the conditioning. With the knot device of Section 4 nothing larger than a cluster or the number of clusters is inverted at any point of the square, the budget of the bridge in [5]. HERC itself inverts nothing, and the edge $\gamma = 0$ adds one inverse per cluster to it.

8 Closing

Until now HERC has been placed with the heuristic methods, as a variant of hierarchical risk parity that budgets by risk contribution. Its budget rule is the global minimum-variance rule. What separates it from the optimum is conditioning, omitted twice, and each omission is a Schur

complement that can be restored by degrees. The dendrogram it builds is used, in the precision reading of its own split formula, only to choose the clusters.

Two things are new in the inner dial. Stevens' identity becomes a one-parameter path with a closed form, a long-only frontier, and, on an equicorrelated cluster, a straight segment whose noise problem is one scalar. And that scalar decides, in closed form, whether estimation error should stop the path short of the cluster's minimum-variance portfolio.

Open directions are the analogues of those in [5]: lost precision and effective damping for the outer dial under a gateway model, global statements for the joint problem, and the non-quadratic risk measures HERC also admits, for which no fitness identity of the form (3) exists and the square has no exact far corner.

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